PROJECT PROPOSAL

Temporary Rental Space Management System of MFU

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BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2026

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PATIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF
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THIS COMPUTER ENGINEERING PROJECT HAS BEEN APPROVED
TO BE PARTIAL FULFILLMENT OF REQUIREMENTS FOR
THE DEGREE OF BACHELOR OF ENGINEERING IN
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Abstract

This senior project proposes the development of a web-based Temporary Rental Space Management System for Mae Fah Luang University. Designed to serve both internal university members and external organizations, the system addresses the inefficiencies of traditional, paper-based reservation methods. By fully digitizing the workflow, the platform aims to eliminate manual processing errors, reduce communication delays, and prevent scheduling conflicts.

The system provides a comprehensive online solution where users can check real-time availability, select customized add-ons, and process secure payments seamlessly. Simultaneously, administrators are equipped with a centralized dashboard for dynamic facility management, digital approval workflows, and data visualization. Ultimately, this system enhances the transparency, convenience, and overall operational efficiency of managing the university's commercial and non-commercial rental spaces.

Keyword: Space Management System, Web Application, Online Booking, Payment Integration, Mae Fah Luang University

CHAPTER 1
INTRODUCTION

1.1 Background and Rational

Mae Fah Luang University provides a variety of facilities and spaces that are frequently utilized for both internal academic activities and external commercial events. Currently, the reservation process relies heavily on manual, paper-based procedures. This traditional approach is highly susceptible to human error, often resulting in scheduling conflicts, delayed communications, and significant complexities in managing financial transactions and official documentation for external clients.

To overcome these administrative bottlenecks, transitioning to an automated, web-based platform is essential. A modern digital reservation system can seamlessly integrate real-time availability tracking, secure payment gateways, and automated status notifications. This not only provides more reliable and user-friendly experience but also supports business logic such as promotional campaigns and service add-ons.

Therefore, this project aims to develop a comprehensive Temporary Rental Space Management System for the university. By replacing outdated manual workflows with robust web architecture, the system will streamline space allocation, improve financial tracking, and equip administrators with data-driven insights to manage the university's resources efficiently.

1.2 Objective

1.2.1 To develop a web-based room reservation system for Mae Fah Luang University

1.2.2 To enable users to view room availability and submit reservation requests online

1.2.3 To provide administrators with tools to manage rooms and reservation requests efficiently

1.3 Scope

1.3.1 System Users

Users are categorized into three distinct authorization levels based on their relationship with Mae Fah Luang University (MFU):

- 1.3.1.1 Internal Users: Individuals operating within the MFU organization.
- 1.3.1.2 External Users: Individuals or entities outside the MFU organization.
- 1.3.1.3 Collaborative Users: Users involved in cross-organizational partnerships or activities.
- 1.3.1.4 System Administrator: Personnel responsible for overseeing facility data, managing reservation requests, and handling administrative workflows.

1.3.2 User Functional Scope:

- Availability Search & Advanced Booking: Check real-time room availability and submit advance reservation requests within specified timeframes.
- Service Customization (Add-ons): Select supplementary services or additional equipment (e.g., specific quantities of tables and chairs) during the booking process, with automated calculation of additional costs.
- Secure Payment Integration: Process transactions securely and seamlessly via an integrated payment gateway.
- Promotions and Incentives: Apply promotional discounts or special offers provided by the system during the reservation process.
- Reservation Management & History: Access personal booking logs, specify reservation purposes, track current statuses, and cancel existing bookings according to system policies.
- Status and Event Notifications: Receive automated alerts regarding booking confirmations, workflow status updates, and announcements for new promotional campaigns.
- Post-Service Feedback: Submit reviews and satisfaction ratings after completing the reservation lifecycle to provide service evaluations.

1.3.3 Administrator Functional Scope:

- Facility & Rate Management: Maintain room records (Add/Edit/Disable) and manage pricing structures. This includes the ability to bulk import room details and room rates directly from spreadsheet datasets (Excel files).
- Administrative Dashboard & Data Visualization: Access a centralized analytics dashboard featuring comprehensive data visualization through interactive graphs and charts. This tool allows for monitoring real-time booking statistics, revenue tracking, and facility utilization, with the ability to drill down into specific data entries for detailed analysis.
- Approval Workflow and Documentation: Process reservation requests by updating statuses (Approve/Disapprove). This includes the ability to import executive authorization files and export official approval documents to serve as verified digital evidence.
- Promotion Management: Create, configure, and manage promotional campaigns to encourage system utilization.
- System Notifications & Broadcasts: Manage automated system alerts and broadcast new promotional information or critical updates to designated user groups.

1.4 Methodology

This project follows an iterative Software Development Life Cycle (SDLC) to ensure the system meets the requirements of Mae Fah Luang University.

- 1.4.1 Requirement Gathering and Analysis: Collect and analyze user requirements by reviewing existing manual reservation processes and examining documents from the Intellectual Property and Innovation Management Division.
- 1.4.2 System Design: Design the system architecture, user interface (UI), and database. This includes designing the relational database schema and mapping data relationships using PostgreSQL and pgAdmin4.
- 1.4.3 System Development: Implement the web application using a modern tech stack. The frontend will be developed using Vue.js and Tailwind CSS for a responsive user interface, while the backend API and server logic will be handled by Node.js.
- 1.4.4 System Testing and Evaluation: Conduct rigorous testing to identify and resolve bugs. This includes testing role-based access for internal and external users, validating the approval workflows, and ensuring the system prevents scheduling conflicts.
- 1.4.5 Documentation and Presentation: Prepare the formal Senior Project 1 and 2 documentation and defend the project before the examination committee.

1.5 Plan

- **Table 1.1** Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 examination										
System development										
System testing and evaluation										
Senior project 2 document preparation										
Senior Project 2 examination										

1.6 Expected Result

1.6.1 A web-based room reservation system for Mae Fah Luang University

1.6.2 Users will be able to reserve rooms conveniently through an online platform

1.6.3 Improved efficiency and accuracy in room reservation management

1.7 Place and Equipment

1.7.1 Equipment

1.7.1.1 Hardware

- Personal computer or laptop
- Internet connection

1.7.1.2 Software

- Frontend: Vue.js, Tailwind CSS
- Backend: Node.js
- Database: PostgreSQL, pgAdmin4

1.7.1.3 Data set

- University room details and specifications

CHAPTER 2
LITERATURE REVIEW

ตารางเปรียบเทียบกับเว็บของห้องในระบบ MFU

CHAPTER 3
ANALYSIS AND DESIGN

ER-Diagram, Use-Case Diagram, User Interface(UI),
System Architecture Design, Swimlane Diagram

CHAPTER 4
CONCLUSION